Definitions of Error and Parameter Estimates.

Keith Lohse, PhD

Assistant Professor

Department of Health, Kinesiology, and Recreation

Department of Physical Therapy and Athletic Training

University of Utah

keith.lohse@health.utah.edu

@keith_lohse

HEALTH
UNIVERSITY OF UTAH

Definition of Errors

- Recall that DATA = MODEL + ERROR…
- Or, for a simple, unconditional model: $y_i = b_0 + e_i$
  - Thus, an error is the difference between our prediction and the observed value, $y_i - b_0 = e_i$.
  - Written this way, we can see that the error is value "left-over" or that "remains to be explained." As such, errors of this type are also referred to as ***residuals***.

Definition of Errors

- But first we have to fit our model parameters.
  - I.e., translate: $y_i = \beta_0 + \varepsilon_i$ into $y_i = b_0 + e_i$

- But what value of b_0 do we want? How do we choose?

  - Recall that b_0 is an estimate of β_0. Intuitively, b_0 first needs to be a good explanation of our data. If we pick a value of b_0 that leads to large errors, then it certainly isn't a good explanation of y_i.

  - More work will need to be done to understand the relationship between b_0 and β_0… but first we want our parameter estimate to be a good description of our data.

Different Methods of Calculating Errors

- Estimating the number of books read over the summer by students.
  - $y_i = \beta_0 + \varepsilon_i$

Books
1
3
5
9
14

Sum of Absolute Errors

- Estimating the number of books read over the summer by students.

- Pessimistic Estimate
  - $y_i = \beta_0 + \varepsilon_i$
  - $y_i = 0 + e_i$

# Books	$\hat{y}_i$	e_i
1	0	1
3	0	3
5	0	5
9	0	9
14	0	14
		32

$\sum |e_i|$

Sum of Absolute Errors

- Estimating the number of books read over the summer by students.

- Optimistic Estimate
  - $y_i = \beta_0 + \varepsilon_i$
  - $y_i = 10 + e_i$

# Books	$\hat{y}_i$	e_i
1	10	-9
3	10	-7
5	10	-5
9	10	-1
14	10	4
		26

$\sum |e_i|$

Sum of Absolute Errors

- Estimating the number of books read over the summer by students.

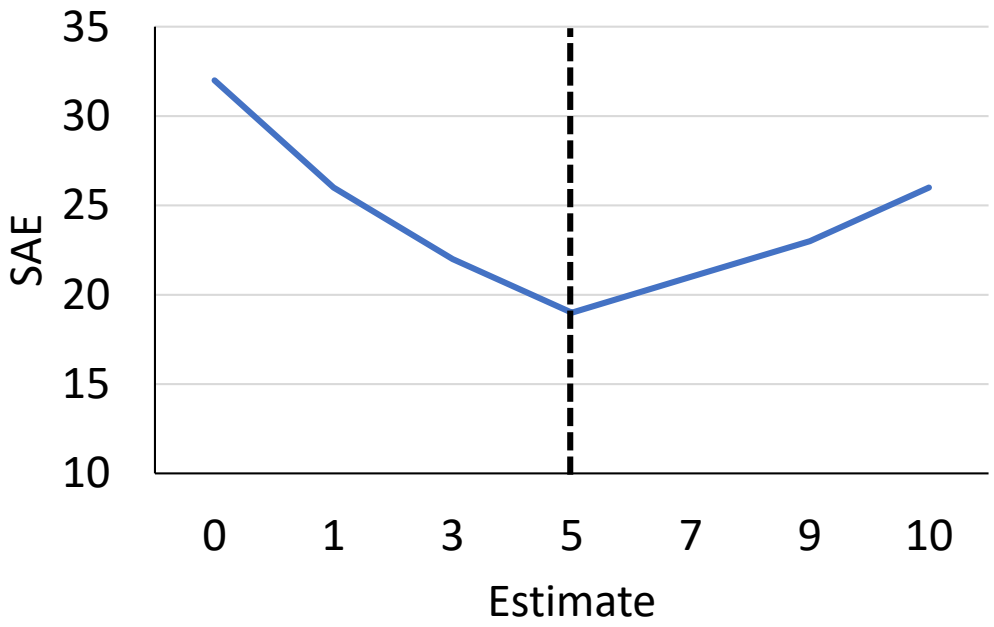
- Moderate Estimate
  - $y_i = \beta_0 + \varepsilon_i$
  - $y_i = 5 + e_i$

# Books	$\hat{y}_i$	e_i
1	5	-4
3	5	-2
5	5	0
9	5	4
14	5	9
		19

$\sum |e_i|$

Sum of Absolute Errors

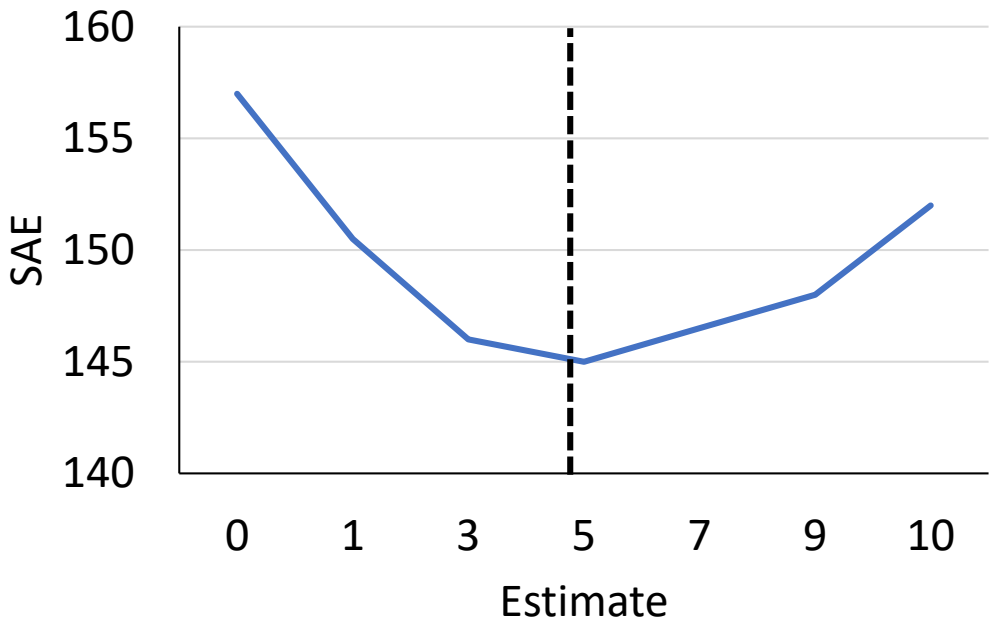
# Books	$\hat{y}_i$	e_i
1	5	-4
3	5	-2
5	5	0
9	5	4
14	5	9
		19



- The median will always produce the smallest sum of absolute errors.

Sum of Absolute Errors

# Books	$\hat{y}_i$	e_i
1	5	-4
3	5	-2
5	5	0
9	5	4
140	**5**	**135**
		145



- The median will ***always*** produce the smallest sum of absolute errors.

Sum of *Squared* Errors

- Estimating the number of books read over the summer by students.

- Pessimistic Estimate
  - $y_i = \beta_0 + \varepsilon_i$
  - $y_i = 0 + e_i$

# Books	$\hat{y}_i$	e_i	e_i^2
1	0	1	1
3	0	3	9
5	0	5	25
9	0	9	81
14	0	14	196
			312

$\sum (y_i - \hat{y}_i)^2$

Sum of *Squared* Errors

- Estimating the number of books read over the summer by students.

- Optimistic Estimate
  - $y_i = \beta_0 + \varepsilon_i$
  - $y_i = 10 + e_i$

# Books	$\hat{y}_i$	e_i	e_i^2
1	10	-9	81
3	10	-7	49
5	10	-5	25
9	10	-1	1
14	10	4	16
			172

$\sum (y_i - \hat{y}_i)^2$

Sum of *Squared* Errors

- Estimating the number of books read over the summer by students.

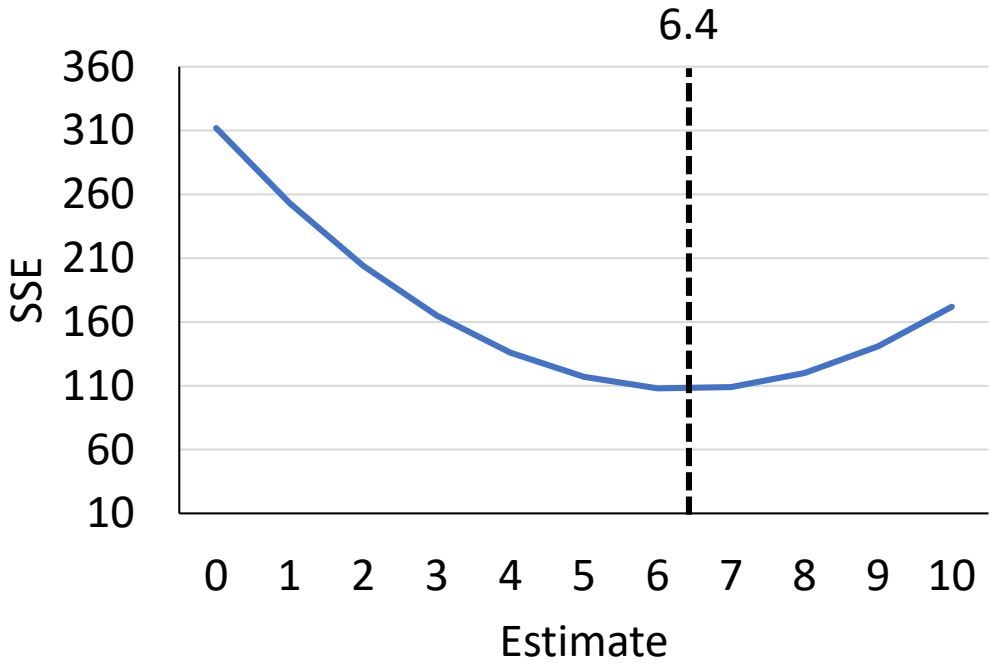
- Moderate Estimate
  - $y_i = \beta_0 + \varepsilon_i$
  - $y_i = 5 + e_i$

# Books	$\hat{y}_i$	e_i	e_i^2
1	5	-4	16
3	5	-2	4
5	5	0	0
9	5	4	16
14	5	9	81
			117

$\sum (y_i - \hat{y}_i)^2$

Sum of *Square* Errors

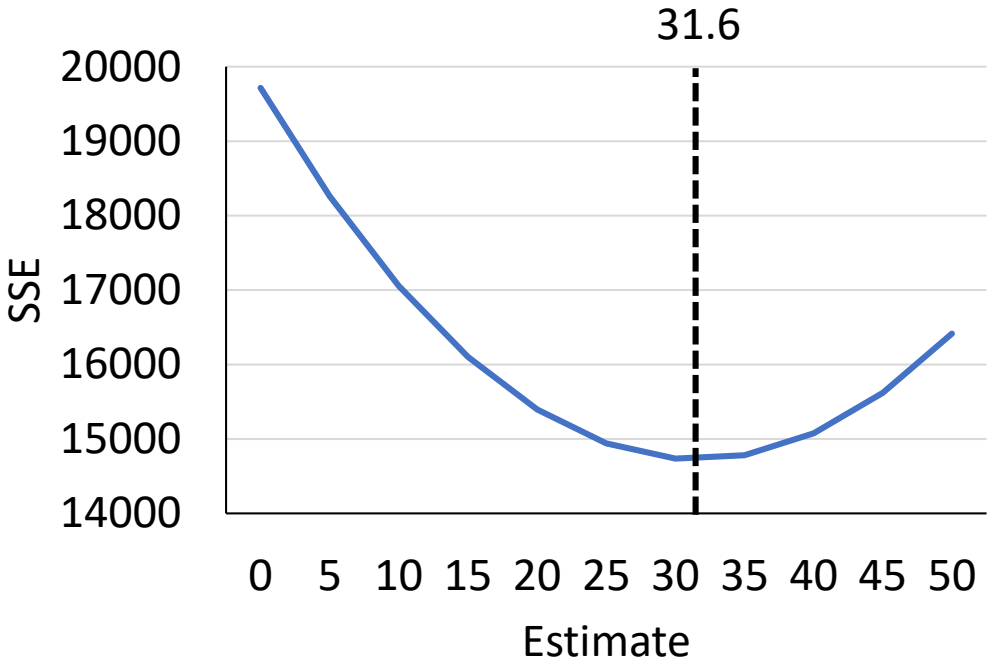
# Books	$\hat{y}_i$	e_i	e_i^2
1	6.4	-5.4	29.16
3	6.4	-3.4	11.56
5	6.4	-1.4	1.96
9	6.4	2.6	6.76
14	6.4	7.6	57.76
			107.2



- The ***mean*** will always produce the smallest sum of ***squared*** errors.

Sum of *Square* Errors

# Books	$\hat{y}_i$	e_i	e_i^2
1	31.6	-30.6	936.36
3	31.6	-28.6	817.96
5	31.6	-26.6	707.56
9	31.6	-22.6	510.76
140	31.6	108.4	11750.56
			14723.2



- The ***mean*** will always produce the smallest sum of ***squared*** errors.
- But also note that the mean is sensitive to outliers, whereas the median is not.

Describing Error

- When describing errors in a sample, ***total*** error is not especially useful, as larger samples will likely produce more error.

- We need to standardize our sums of errors with respect to the sample size.
  - For SAE, we will take median absolute error.

  - For SSE, we will take the mean squared-error (a *variance*) or the root mean squared-error (a *standard deviation*).

$$MSE = \frac{SSE}{n-p} = \frac{\sum(y_i - \hat{y}_i)^2}{n-p}$$

$$RMSE = \sqrt{MSE} = \sqrt{\frac{\sum(y_i - \hat{y}_i)^2}{n-p}}$$

[Where n = number of observations and p = number of ***estimated*** parameters]

Describing Error

- Thus if $y_i = \beta_0 + \varepsilon_i$ we are ***<u>estimating</u>*** one parameter (β_0), so $n - p$ is really $n - 1$.

$$RMSE = \sqrt{MSE} = \sqrt{\frac{\sum(y_i - \hat{y}_i)^2}{n-1}}$$

- Additionally, the besting fitting estimate (b_0) will be the mean of all of the observations ($\bar{y}$). Thus, for each person we predict the mean and can write:

$$RMSE = \sqrt{MSE} = \sqrt{\frac{\sum(y_i - \bar{y})^2}{n-1}}$$

[Which is the formula for the standard deviation in a sample you have probably encounter before.]